

serve the scanning at TP3 through TP7 using an oscilloscope. The serial data for shift resistors can be observed at TP2.

Software

The source program for the microcontrollers is written in 'C' language and compiled using Keil μ Vision4 compiler. The generated hex code is burnt into the microcontrollers by using a suitable programmer.

GSM section. The commands used to configure the GSM module are:

1. **AT+CRES.** This command is used to reset the GSM module.
2. **AT+CMGF=1.** This command is used for setting up the GSM module to send data in ASCII format.
3. **AT+CNMI=2,2,0,0,0.** This command is used to get complete message on the serial port.

Functions used in the program of the GSM section are:

1. **receive()**. Receives serial data from the GSM module and saves it to a predefined location

PARTS LIST

Semiconductors:	
IC1-IC8	- 74HC595 shift register
IC9, IC10	- P89V51RD2 microcontroller
IC11	- MAX232 RS232 driver
IC12	- 78S05, 5V regulator
T1-T5	- IRF9630 p-channel MOSFET
D1-D4	- 1N5400 rectifier diode
LED1	- 5mm LED
DIS1-DIS8	- TA20-11 EWA 5x7 common-anode dot-matrix display
GSM modem	
	- SIM300
Resistors (all $\frac{1}{4}$-watt, $\pm 5\%$ carbon):	
R1-R7	- 10-kilo-ohm
R8	- 330-ohm
Capacitors:	
C1, C2, C5, C6	- 33pF ceramic disk
C3, C4, C7-C11	- 1 μ F, 16V electrolytic
C12	- 1000 μ F, 35V electrolytic
C13	- 0.1 μ F ceramic disk
Miscellaneous:	
X1	- 230V AC primary to 12V, 3A secondary transformer
	- Push-to-on tactile switch
S1, S2	- 11.0592MHz crystal
X _{ALP} , X _{AL2}	- DB9 com port connector

2. **msdelay()**. Provides delay
3. **Transmit_data()**. Sends serial data to the display section
4. **Gsmcmdsend()**. Sends GSM commands to the GSM module

Display section. Functions used in

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com.

the program of the display section:

1. **datazero()**. Clears the data array
2. **clock1()**. Provides clock for serial shift register
3. **data latch()**. Provides latch signal for shift register
4. **msdelay()**. Provides delay
5. **dash()**. Shifts data in the shift register
6. **receive()**. Receives serial data from the GSM section
7. **default_msg()**. Displays default message "NO Msg"
8. **data_compare()**. Compares the data with character set
9. **data_insert()**. Inserts serial data
10. **dot_display()**. Displays the message

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